

Supplementary material: HCA-DSE

Full budget sweeps and validation subsets accompanying the main manuscript. All values are generated from the same result CSVs as the main-text summaries.

Table S1: Full budget-matched comparison on S2. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.059
		random ($B=100$)	100	0.037	0.520	0.2030	0.001
		random ($B=250$)	250	0.037	0.730	0.1244	0.002
		random ($B=500$)	500	0.111	0.853	0.0766	0.005
		random ($B=1000$)	1000	0.185	0.899	0.0540	0.011
		NSGA-II ($B=100$)	100	0.000	0.569	0.1889	0.013
		NSGA-II ($B=250$)	250	0.204	0.868	0.0519	0.039
		NSGA-II ($B=500$)	500	0.667	0.981	0.0101	0.091
		NSGA-II ($B=1000$)	1000	0.907	0.999	0.0005	0.207
		HCA-DSE	48	1.000	1.000	0.0000	0.044
S2	control	exhaustive	3 888	1.000	1.000	0.0000	0.056
		random ($B=100$)	100	0.000	0.522	0.2114	0.001
		random ($B=250$)	250	0.040	0.728	0.1248	0.002
		random ($B=500$)	500	0.120	0.846	0.0776	0.005
		random ($B=1000$)	1000	0.160	0.897	0.0530	0.012
		NSGA-II ($B=100$)	100	0.000	0.565	0.1573	0.014
		NSGA-II ($B=250$)	250	0.280	0.900	0.0398	0.041
		NSGA-II ($B=500$)	500	0.680	0.993	0.0065	0.095
		NSGA-II ($B=1000$)	1000	0.780	0.999	0.0030	0.234
		HCA-DSE	38	1.000	1.000	0.0000	0.044
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.041
		random ($B=100$)	100	0.000	0.201	0.3725	0.001
		random ($B=250$)	250	0.062	0.579	0.1720	0.002
		random ($B=500$)	500	0.125	0.804	0.0944	0.004
		random ($B=1000$)	1000	0.188	0.868	0.0732	0.009
		NSGA-II ($B=100$)	100	0.000	0.269	0.3144	0.014
		NSGA-II ($B=250$)	250	0.250	0.723	0.0983	0.046
		NSGA-II ($B=500$)	500	0.562	0.937	0.0224	0.110
		NSGA-II ($B=1000$)	1000	0.875	1.000	0.0017	0.270
		HCA-DSE	26	1.000	1.000	0.0000	0.024

Table S2: Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.372
		random ($B=100$)	100	0.000	0.324	0.4165	0.001
		random ($B=250$)	250	0.000	0.403	0.2781	0.002
		random ($B=500$)	500	0.000	0.605	0.2188	0.004
		random ($B=1000$)	1000	0.000	0.653	0.1500	0.009
		NSGA-II ($B=100$)	100	0.000	0.069	0.5304	0.012
		NSGA-II ($B=250$)	250	0.000	0.294	0.2763	0.041
		NSGA-II ($B=500$)	500	0.258	0.754	0.1242	0.094
		NSGA-II ($B=1000$)	1000	0.500	0.945	0.0466	0.235
		HCA-DSE	54	1.000	1.000	0.0000	0.156
S3	control	exhaustive	34 560	1.000	1.000	0.0000	0.349
		random ($B=100$)	100	0.000	0.312	0.4466	0.001
		random ($B=250$)	250	0.000	0.377	0.3157	0.002
		random ($B=500$)	500	0.000	0.579	0.2286	0.004
		random ($B=1000$)	1000	0.000	0.652	0.1488	0.009
		NSGA-II ($B=100$)	100	0.000	0.114	0.4307	0.012
		NSGA-II ($B=250$)	250	0.037	0.539	0.2269	0.041
		NSGA-II ($B=500$)	500	0.315	0.877	0.0456	0.099
		NSGA-II ($B=1000$)	1000	0.648	0.983	0.0236	0.237
		HCA-DSE	41	1.000	1.000	0.0000	0.170
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.384
		random ($B=100$)	100	0.000	0.045	0.5926	0.001
		random ($B=250$)	250	0.000	0.224	0.4782	0.002
		random ($B=500$)	500	0.000	0.361	0.2993	0.004
		random ($B=1000$)	1000	0.000	0.476	0.2061	0.009
		NSGA-II ($B=100$)	100	0.000	0.023	0.7297	0.014
		NSGA-II ($B=250$)	250	0.000	0.126	0.3366	0.040
		NSGA-II ($B=500$)	500	0.185	0.682	0.2273	0.101
		NSGA-II ($B=1000$)	1000	0.741	0.944	0.1677	0.259
		HCA-DSE	43	1.000	1.000	0.0000	0.107

Table S3: Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.0687	0.3242	0.998	-0.545	0.0	54
telemetry	250	0.2941	0.4025	0.896	-0.23	0.0	54
telemetry	500	0.7539	0.6053	0.338	0.08	0.258	54
telemetry	1000	0.9448	0.6535	0.299	0.1	0.5	54
control	100	0.1144	0.3124	0.973	-0.355	0.0	41
control	250	0.5391	0.3772	0.462	0.02	0.037	41
control	500	0.8767	0.5795	0.082	0.26	0.315	41
control	1000	0.983	0.6524	0.054	0.3	0.648	41
sensing	100	0.0226	0.0449	0.965	-0.333	0.0	43
sensing	250	0.126	0.224	0.548	-0.02	0.0	43
sensing	500	0.6823	0.3609	0.205	0.155	0.185	43
sensing	1000	0.9439	0.4763	0.142	0.2	0.741	43

Table S4: Full testbed validation, including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as –.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat spread [ms]
telemetry (all)	12	20.3	7.54	0.839	0.727	0.852	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	16.2	2.91	0.829	0.733	0.818	1.0	0.18
control (all)	12	15.7	2.56	0.872	0.779	0.917	0.909	0.27
control ($\rho \leq 0.75$)	10	9.5	2.4	0.778	0.674	0.852	0.833	0.17
sensing (all)	12	50.0	20.0	0.949	0.87	0.957	0.957	2.17
sensing ($\rho \leq 0.75$)	10	49.7	19.28	0.985	0.944	1.0	1.0	1.86
sensing (reproducible)	8	41.7	18.77	0.994	0.982	1.0	–	1.49
all workloads (all)	36	28.7	7.54	0.97	0.893	0.968	0.982	0.49
all workloads ($\rho \leq 0.75$)	26	26.5	3.91	0.948	0.861	0.962	0.982	0.4
all workloads (reproducible)	32	23.9	4.54	0.96	0.881	0.963	0.979	0.38